Architecture and Calibration of a Multi-channel Electrical Impedance Myographer

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Abstract. Electrical impedance myography (EIM) is a non-invasive measurement technique capable of determining physiological and morphological changes in skeletal muscle tissue. The method consists of applying an alternating current on the skin through two electrodes to determine the bioimpedance of the medium. EIM has contributed to the study of degenerative diseases that changes the electrical properties of muscle tissue during its evolution of the disease. The objective of this work is to develop a portable and low-cost impedance measurement instrument to assist in activities related to the study and diagnosis of muscle diseases. We present a modular hardware architecture that allows the impedance measurement of multiple muscles at the same time, but in the present work we test the performance of the prototype with a single impedance measurement channel implemented. The tests were performed with known theoretical impedance calculated to evaluate the accuracy of the impedance meter, as well as other performance tests of the system.

Keywords: Electrical impedance myography · Instrumentation · Hardware design

1 Introduction

Bioimpedance can be defined as the opposition to the passage of electrical current in biological tissues. The smallest building block of life is the cell that in a living tissue is filled and surrounded by fluids containing diluted ions necessary for metabolism. The cell can be modeled mathematically by an association of resistors, representing internal and external fluids, and by capacitors representing the lipid layers of cell membranes and organelles [9]. In this way, biological tissue is not a purely resistive medium to the passage of alternating current and has capacitive reactance that is dependent on frequency. Therefore, more vascularized tissues such as muscles contain a greater amount of ions in their structure making them good conductors of current, on the other hand, less capillaryized bone and fat tissues are poor conductors. Therefore, it is possible to distinguish

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the different biological tissues by determining the impedance of the region of interest. The advantage of this measurement technique is that it reduces the costs of biomedical equipment used for monitoring and diagnosing patients.

There is an increasing number of researches associated with the study of the electrical properties of biological tissues, such as muscle dynamics and physiology [17]; degenerative muscle diseases such as amyotrophic lateral sclerosis [4,15] and muscle fatigue related to sports activities [2,5]. Due to the large number of researches interested in the analysis of the electrical properties of biological tissues, there is a demand for measuring instruments to meet this need.

Some commercial instruments for impedance measurement have been used in these researches. The mScan^{TM} EIM manufactured by the Myolex company [13] is a portable single-channel measuring instrument with a 50 kHz biological tissue excitation current. Another equipment used in the research is the Quantum V Segmental BIA from the company RJL Systems [14], which has 12 multiplexed measurement channels with an excitation frequency also of 50 kHz. Both equipments are designed to measure arm, leg, torso muscles, including the upper and lower regions of the human body.

Although this work presents the development of the concept of a single measurement channel, we adopted a modular design criterion with the aim of developing in the future an EIM hardware with multiple measurement channels to study different muscle groups simultaneously. Thus, this project aims to build a portable and autonomous equipment to measure the bioimpedance of skeletal muscle tissue using electrodes installed on the patient's skin with a excitation current 31.5 kHz.

2 Background

Electrical impedance myography can be used to study skeletal muscle through the electrical properties of this tissue [16]. The method consists of applying a sinusoidal current excitation signal of known amplitude and frequency through two electrodes in the region of interest, while the electrical voltage generated by this current is measured in the same electrodes. Once the current and voltage are known, it is possible to determine the bioimpedance of the medium using Ohm's Law. Thus, EIM has the potential to study the evolution of neuromuscular degenerative diseases by analyzing changes in muscle tissue bioimpedance.

The Fig. 1 shows an electronic model of the cell tissue [9] composed of the resistivity of the extra-cellular fluid R_e; by the resistance R_m and capacitance C_m of the lipid layer of the membrane; and the resistivity of the fluids and organelles inside the cell with resistance R_i. The equivalent electronic circuit of the model can be reduced according to the Fig. 1 and represented by the Eq. 1, where Z_{eq} is the total equivalent impedance; $Z_e = R_e + j0$ is the external fluids impedance; $Z_i = R_i + j0$ is the internal fluid impedance and Z_m is total cell membrane impedance representing the parallel association of resistance R_m and capacitance jX_{C_m}, according to Eq. 2.

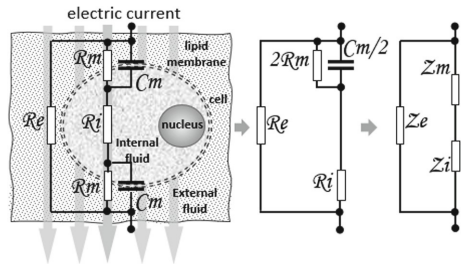


Fig. 1. Equivalent cell model, adapted from [9].

$$Z_{eq} = \left(\frac{1}{Z_e} + \frac{1}{Z_i + Z_m} \right)^{-1} \tag{1}$$

$$Z_m = \left(\frac{1}{2R_m} + j\frac{2}{X_{C_m}} \right)^{-1} \tag{2}$$

The excitation signal of the sinusoidal electric current is typically applied at frequencies that vary from 10 Hz to a few MHz in order to adjust the frequency response according to the object under study. At low frequencies, the reactive component of the impedance, due to the capacitance of the muscle cell membrane tends to be much larger (open circuit) than the resistive component. Therefore, the current bypasses the cellular membrane and flows mostly in the extracellular fluid. At high frequencies the capacitive reactance becomes much smaller (short-circuit) than the resistive component inside the cell, increasing the current intensity in the intracellular fluid. Once the amplitude, frequency and phase of the applied excitation signal are known, it is possible to determine the impedance of the muscle by analyzing changes in the amplitude and phase of the voltage and current measured in the region of interest.

With the EIM it is possible to analyze the impedance changes in healthy muscle tissue by determining the impedance distributed in the longitudinal and transverse axes. Thus, the progression or regression of muscle diseases such as amyotrophic lateral sclerosis, injuries, fat rate, etc. can be monitored [10,15] by analyzing the changes in muscle tissue impedance.

3 Proposed Architecture

3.1 Overview

The design approach adopted was to develop the equipment in electronic modules where each acquisition channel measures an analog voltage signal. This concept allows simultaneous impedance measurement of N samples through $N + 1$ modules, being that N channels are responsible to measure the voltages and one channel to indirectly determines the current by measuring the voltage across a sentinel resistor of known value.

A single voltage-controlled current source individually excites all samples by an analog multiplexer. The maximum number of measurement channels is related to the model of the multiplexer installed in the system. The sinusoidal excitation signal from the current source can be generated either by an internal circuit or by an external signal generator.

The electrode signal is conditioned and filtered in the first stages of the module. The objective is to reduce 50/60 Hz noise from the power supply and also noise with frequencies above the frequency band of interest. Then, the conditioned signal is sent to the next stage composed by an analog-to-digital converter (ADC) circuit. The ADC is the AD7762 integrated circuit manufactured by the company Analog Devices®, whose main characteristics are: (i) differential input of the modulator; (ii) 24-bit resolution; (iii) sampling frequency of 625 kSPS (kilo Samples Per Second) and (iv) signal-to-noise ratio (SNR) of 106 dB [3]. The microcontroller of the channel acquires the sample data discretized by the ADC. The microcontroller used is the 32-bit Atmel SAM3X8E ARM Cortex-M3 which is embedded into the Arduino Due open source platform. The functional block diagram of the Impedance Meter is shown in the Fig. 2.

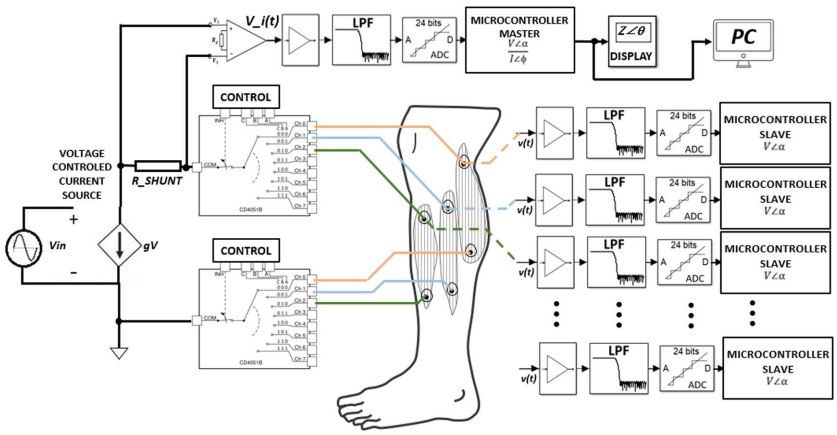


Fig. 2. Functional block diagram of the impedance meter with multiple channels

The design approach shown in Fig. 2 is convenient to implement an impedance meter with any number of simultaneous measuring channels depending only on the choice of the number of channels of the mux integrated circuit. In the present work we have implemented only one impedance measuring channel as a proof of concept. The following subsections detail the implementation of this prototype.

3.2 Acquisition Channels

Hardware Description. The constructed impedance meter prototype have two different types of measurement channels: one called the master channel that

is responsible to measure the current and the other is called slave channel responsible to measure the voltage. The hardware of master and slave channels are similar, the difference between them is the addition of an IC INA129 instrumentation amplifier circuit in the master channel. It is used to amplify the low voltage across the sentinel resistor (Fig. 3). A first order high pass filter (HPF) with 150 Hz of cutoff frequency is placed between the terminals of the sentinel resistor and the inputs V^+ and V^- of the preamplifier INA129 in order to eliminate any DC component from the previous stage.

Notice that the hardware of the master and slave channels is identical except for the presence of the preamplifier on the master channel. Since the PCB of both circuits are the same, we use a jumper to bypass the leads of the preamplifier.

The next stage consists of a first order low pass filter with cutoff frequency of 300 kHz to reduce high frequency noise. In sequence, the next stage is a single-ended to differential signal converter circuit. This circuit is based on a unity gain inverting amplifier with one input and two outputs V_{in}^+ and V_{in}^-, where V_{in}^+ is the original input and V_{in}^- is the input shifted by 180 °C. These outputs will serve as inputs of the ADC. Since the AD7762 ADC has differential inputs, so this adaption circuit through the AD8021 operational amplifier is mandatory [3].

To determine the current it is necessary to measure the voltage drop across the sentinel resistor ($R_{SHUNT} = 56\,\Omega$). Since this voltage drop is small, it is necessary to introduce a preamplifier. The implemented current source supplies from 1 to 10 mA, then the maximum voltage over R_{SHUNT} is only 560 mV. Therefore, the preamplifier multiply this voltage to optimize the input range of ADC which is ±3.6 V. In the implemented circuit, the signal is amplified with a gain factor of 5 through the instrumentation operational amplifier INA129 [7] as seen in the Fig. 3. It is worth noting that only the master channel needs this circuit.

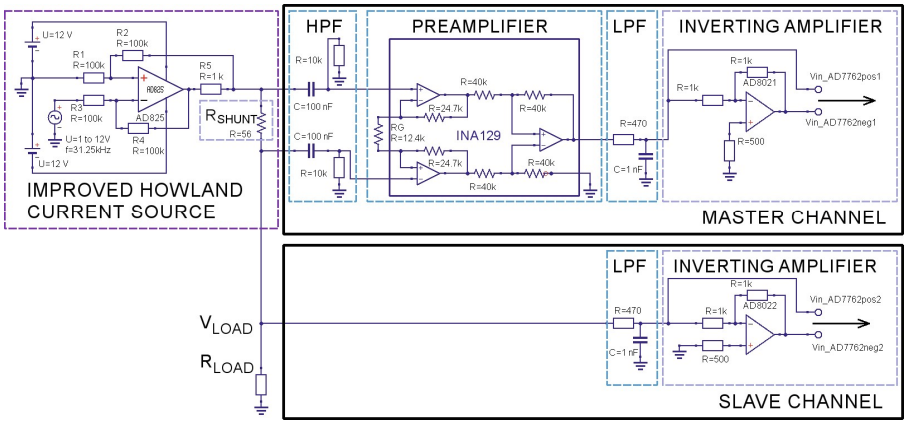


Fig. 3. Electronic schematic of Howland current source and Master/Slave channels signal Conditioners.

Software Description. Regarding the algorithm, the IDE (Integrated Development Environment) of the Arduino microcontroller is divided into two parts. In the first part `void setup()` are inserted the system initialization and configuration instructions that will be executed only once. In the second part `void loop()` are inserted the instructions that are executed continuously by the main program.

First, in `void setup()`, it is necessary to configure the ADC, which remains turned off in default mode, sending the commands that will determine the component's operation settings to Control Registers 1 and 2, such as: power-on; sampling frequency; clock division factor; filter configuration, etc. At this stage, the microcontroller sends 16-bit address words to Control Registers 1 and 2 signaling that it will write data to it. AD7762 understands that the word following the address is a data control word and stored it in the respective register [3]. It is worth noting that the sampling frequency of 312.5 kHz is determined at this stage. Then, still at this stage, the master microcontroller sends data to initialize the display, informs the operator that the equipment is ready to perform an impedance measurement and finishes the setup procedure.

In `void loop()`, once a measurement is triggered, the microcontroller reads two 16-bit words from the sample acquired by the ADC at that instant. This data is rearranged into a single 24-bit word and stored in memory. This digital word contains the 16 bits of the Most Significant Word (MSW) and the 8 bits of the Least Significant Word (LSW) of the discretized sample value. The microcontrollers - master and slaves - reads 10 samples for each period of the sinusoidal signal and exits looping. Next, it determines the amplitude and phase angle of the voltage and current, respectively, using the quadrature demodulation process [19] (see details in Appendix A). The microcontroller of the master channel reads the data from the ADC_{master} and then, it requests the slave channel to send the amplitude and phase of the voltage over the load $V_{load}\angle\phi^\circ$ calculated in that channel. With these data the master channel calculates the impedance $Z_{load}\angle\theta^\circ$ by dividing the voltage of the load by the current and sends the value to a LCD display or microcomputer. Communications between the microcontroller of the master channel and the slave channels are carried out by the I^2C (Inter-Integrated Circuit) protocol. In addition, each channel can connect with the personal computer via the USB (Universal Serial Bus) port.

3.3 Analog-to-Digital Converter

The electronic circuit of the analog acquisition channel has as main component the integrated circuit AD7762. As mentioned in Sect. 3.1, it is a 24-bit resolution analog-to-digital converter capable of data acquisition at 625 kSPS and signal-to-noise ratio (SNR) of the order of 106 dB. This is a high-performance ADC designed under the Σ-Δ (sigma-delta) architecture where samples are oversampled at an acquisition rate much higher than the Nyquist frequency. The sample discretizes the 24 bits in two steps, or rather, in two 16-bit words. When the ADC finishes the discretized conversion of a sample, the DRDY (Data Ready Output - pin 38) control signal of the ADC goes to logic level '0' indicating that

the most significant 16 bits of the word (MSW) are available for reading in the buffer.

To perform the reading, the microcontroller must send a signal of logic level '0' to the CS (Chip Select - pin 40) and RD/WD (Read/Write - pin 39) pins, respectively, for at least 6 cycles of clock and acquire the 16 bits MSW from word. After this time interval, the microcontroller resets logic level '1' on the CS and RD/WR control pins for the same period of 6 cycles, in this way, the ADC provides the second part of the word with the 8 least significant bits LSW of the converted signal, along with 8 bits of the STATUS of the conversion. Again, the microcontroller sends logic level 0 for 6 cycles to the CS and DR/WR control pins to read the second part of the word, returning to logic level '1' after reading. Note that the DRDY signal goes to logic level 0 only once for each valid 24-bit data. The microcontroller then rearranges the 16 MSW bits of the first part of the word with the 8 most significant bits of the second part (LSW) to store the signal discretization in just a 24-bit word. The 8 least significant bits of the LSW word present the status of the discretization performed, and each bit informs a certain condition, for example: DVALID (valid data); OVR (overload), among others. Once the reading cycle is completed, the microcontroller waits for signaling from the DRDY pin to read a new sample.

3.4 Current Source

The current source used is built around the design known as the Discrete Improved Howland Current Pump described by TEXAS INSTRUMENTS [6,8]. This type of current source is highly used in bioimpedance analysis applications due to its high output impedance and has a constant amplitude over a wide range of frequency [1]. It uses just one op-amp and five resistors to supply and control the current in the load resistor R_{LOAD} as seen in the Fig. 3. The load current I_{LOAD} can be calculated using Eq. 3 and the circuit gain G is given by Eq. 4.

$$I_{LOAD} = \frac{G(V_p - V_n)}{R_5} \tag{3}$$

$$G = \frac{R_2}{R_1} = \frac{R_4 + R_5}{R_3} \tag{4}$$

The current source was tested to verify its performance over different loads. This test was initially performed only with resistive loads to determine the current source capacity as a function of the load. The test consisted of injecting the current supplied by the source in a determined load resistor and measuring the voltage drop with an oscilloscope Minipa model MVB-DSO de 100 MHz [11]. The actual values of the resistors were obtained using a digital multimeter Tektronix model TEK DMM 830 [18]. In the test, the current source excitation signal is generated by the function signal generator model Minipa MFG-4202 [12]. Figure 4 presents the experimental survey of the current delivered by the source as a function of load for the range of interest from zero to 1 kΩ. Analyzing

the behavior test data, we verified that the current source enters the non-linear region for resistance values above 5 kΩ.

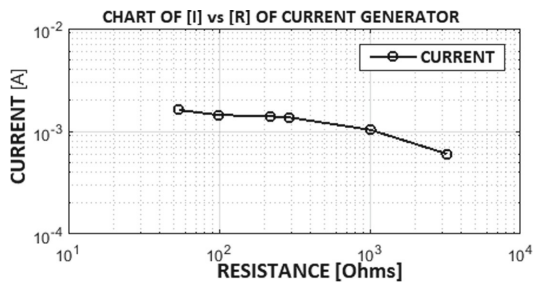


Fig. 4. Experimental test of current source and resistive load.

4 Testing Protocols

In the following subsections we describe some tests that were performed to evaluate the performance of the impedance meter after the construction of the first prototype was finished.

4.1 Sample Rate Performance Test

The AD7762 can be programmed to operate at four different sampling frequencies depending on the desired application: 625 kSPS, 312.5 kSPS, 15.75 kSPS and 7.875 kSPS. However, as demonstrated in Sect. 3.3, the microcontroller has to read two 16-bit words on the ADC data bus, interpret this information and store the discretized sample value in a 24-bit word. For the fastest sampling rate of 625 kSPS the microcontroller has 1.6 µs to accomplish this task. Since the microcontroller chosen is the Arduino Due, a test was performed with each of these sampling frequencies and voltage of 1 V peak amplitude to verify the performance of the ADC-Arduino combination. In this test, we verified that the Arduino does not behave properly only for the sampling frequency of 625 kSPS, without prejudice to the other frequencies. Data were stored on the computer in text mode. In this step, we verified that the Arduino Due microcontroller presented the best performance of discretization of input signals for a sampling rate of 312.5 kSPS. Therefore, we adopted this sampling frequency for the operation of the system.

4.2 Excitation Frequency Performance Test

In the next step, analog signals generated with the Minipa generator were acquired. Four signals were generated to perform the tests with the amplitudes in the values of 150 mV_p, 500 mV_p, 1 V_p and 1.6 V_p, respectively. The sampling

rate was 312.5 kHz for all signals. The signals were generated at the frequencies of 10 kHz, 31.25 kHz, 50 kHz and 125 kHz totalizing 16 different measurements. The graphs in the Fig. 5 show an example of these measurements for the voltage of 500 mV_p in the range of 100 µs. Analyzing these data, we decided to adopt the frequency of 31.25 kHz for the excitation of the sample under test and to acquire ten points per sinusoidal period since the sampling frequency is 312.5 kHz.

The stored signals were analyzed in the MATLAB® software to determine the total harmonic distortion rate (THD) and the FFT of the discretized signals. The THD varied between 1.01% and 3.7% increasing distortion with increasing amplitude. The THD was calculated in the Matlab program using the function $[r, harmpow, harmfreq] = thd(x, fs, n)$ where: r is the value of the THD in dB; $harmpow$ is the harmonic power value; $harmfreq$ is the harmonic frequency; thd is the function that calculates the THD; x is the name of the file with the discretized data; fs is the sampling frequency; and n and the desired harmonic iteration number. It is worth noting that the THD also depends on the quality of the signal generator to maintain the monotonicity of the sine wave. In the case of the Minipa MFG-4202 function generator, the manual [12] informs that the sine wave distortion is less than 2% (10 kHz, 5 Vpp). Figure 6 present an example demonstrating these analyses.

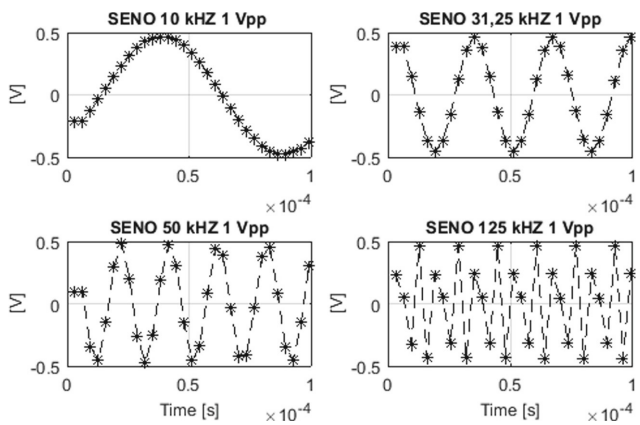


Fig. 5. Example of 4 discretized samples - sample Rate 312.5 kHz.

4.3 Impedance Accuracy Test

Subsequently, measurements of previously known impedances were performed. The impedances were produced through associations in parallel between capacitors and resistors. The actual values of the components were measured with the multimeter model TEK DMM 830 and these data were used to theoretically determine the impedances through the Eqs. 5. A theoretical table with 28 different impedances was created using the values of the real components, with

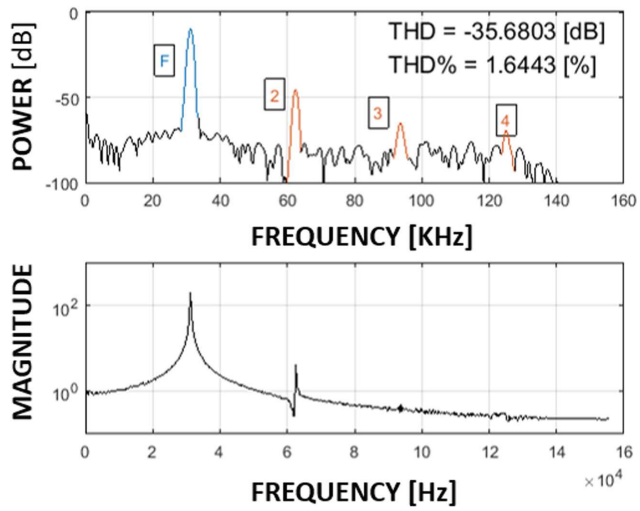


Fig. 6. THD sinusoidal signal 500 mVp 31.23 kHz. - sample rate 312.5 kHz.

associations in parallel and an excitation frequency of 31.25 kHz. The calculation of the resulting impedance of a resistor R in parallel with capacitor with reactance X is given by Eq. 5:

$$Z = \frac{jRX}{R + jX} = \frac{RX^2}{R^2 + X^2} + j\frac{R^2 X}{R^2 + X^2} \ , \tag{5}$$

In the test, we deliver to the load the highest possible current for the impedance in the analysis. The amplitudes of the excitation currents varied from 1 to 10 mA, and the ratio of the amplitudes of the signals sent to the inputs of the ADC's is the highest possible, reducing discretization errors.

The test is performed with the help of the oscilloscope Minipa MVB-DSO and the function generator Minipa MFG-4202. Since the current is controlled as a function of the voltage drop across the sentinel resistor, we install the oscilloscope probe across the terminals of this resistor. Then, through the generator, a low-amplitude voltage is applied to the input of the current source and the voltage amplitude (V_{in}) is gradually increased until reaching the best relationship between voltage and sentinel resistor. Then, measurement tests were performed having as standard reference the impedances calculated according to the Eq. 5. Resistors with values 55.1 Ω, 100.0 Ω, 220.0 Ω, 292.4 Ω, 497.0 Ω, 668.0 Ω and 994.0 Ω were used. The resistors were associated in parallel with each of the following capacitors: $C_1 = 1.5$ nF, $C_2 = 3.3$ nF and $C_3 = 4.4$ nF. The Table 1 presents a summary of the actual impedance measurements performed with the impedance meter [M] and its respective theoretical [T] value calculated.

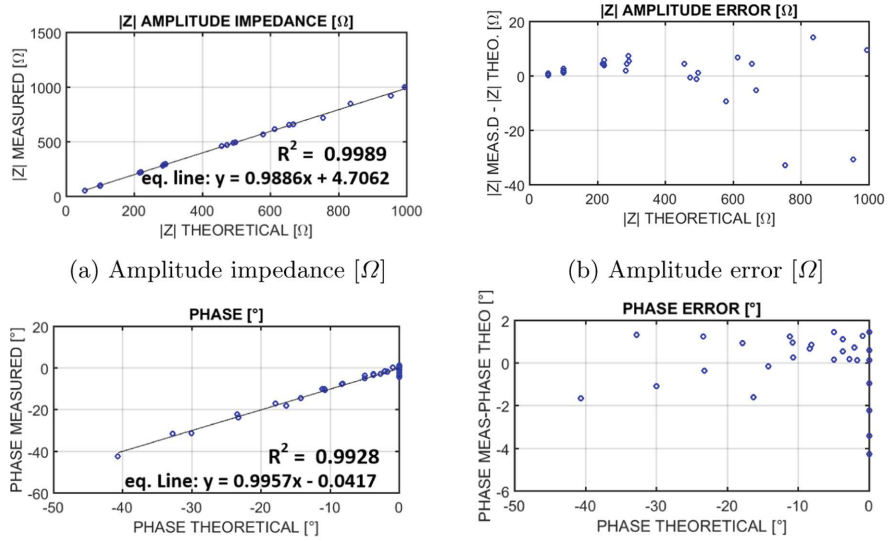
Table 1. Difference between the theoretical impedance, marked as [**T**] and the measured impedance, marked as [**M**]. The second column show a purely resistive load with resistor R while third to fifth column shows a capacitive load with R in parallel with capacitors $C_1 = 1.5$ nF, $C_2 = 3.3$ nF and $C_3 = 4.4$ nF.

		Impedance Association			
		R [Ω]	$R\|C_1$ [Ω]	$R\|C_2$ [Ω]	$R\|C_3$ [Ω]
* $R = 55.1\,\Omega$	**T**	$55.1\angle 0.0°$	$55.1\angle -0.9°$	$55.1\angle -2.0°$	$55.0\angle -2.7°$
	M	$56.1\angle 1.5°$	$56.1\angle 0.4°$	$55.3\angle -1.3°$	$55.8\angle -2.5°$
* $R = 100.0\,\Omega$	**T**	$100.0\angle 0.0°$	$99.9\angle -1.7°$	$99.7\angle -3.7°$	$99.6\angle -4.9°$
	M	$102.0\angle 0.6°$	$102.7\angle -1.5°$	$101.3\angle -3.1°$	$100.8\angle -3.5°$
* $R = 220.0\,\Omega$	**T**	$220.0\angle 0.0°$	$219.5\angle -3.7°$	$217.8\angle -8.1°$	$216.1\angle -10.8°$
	M	$225.9\angle 0.1°$	$223.5\angle -2.6°$	$222.3\angle -7.3°$	$220.6\angle -9.8°$
* $R = 292.4\,\Omega$	**T**	$292.4\angle 0.0°$	$291.3\angle -4.9°$	$287.3\angle -10.7°$	$283.5\angle -14.2°$
	M	$297.9\angle -0.9°$	$298.8\angle -4.7°$	$291.8\angle -10.4°$	$285.6\angle -14.3°$
* $R = 497.0\,\Omega$	**T**	$497.0\angle 0.0°$	$491.8\angle -8.3°$	$473.1\angle -17.9°$	$456.7\angle -23.2°$
	M	$498.3\angle -2.2°$	$490.6\angle -7.6°$	$472.6\angle -16.9°$	$461.2\angle -23.6°$
* $R = 668.0\,\Omega$	**T**	$668.0\angle 0.0°$	$655.4\angle -11.1°$	$613.0\angle -23.4°$	$578.6\angle -30.0°$
	M	$662.8\angle -3.4°$	$660.1\angle -9.9°$	$619.9\angle -22.1°$	$569.4\angle -31.1°$
* $R = 994.0\,\Omega$	**T**	$994.0\angle 0.0°$	$953.9\angle -16.3°$	$835.7\angle -32.8°$	$741.1\angle -40.7°$
	M	$1003.6\angle -4.3°$	$923.4\angle -17.9°$	$849.9\angle -31.5°$	$721.4\angle -42.3°$

Each sample measured in the Table 1 is the trimmed mean of 100 impedance measurements acquired in the tests. In this procedure, the upper and lower end quartiles of the sample set were eliminated to calculate the trimmed mean. The quartile was calculated as follows. As these are complex numbers, firstly, the modules were sorted, which have greater weight in the polar form, where 12.5% of the samples at the upper and lower ends were eliminated at this stage, resulting in a new data set. In this reduced data set, a new sort was performed, only this time the angles were sorted, and again 12.5% of the upper and lower samples were eliminated, resulting in a set with 50 valid data in polar form. Then, these data were converted from polar to rectangular form to calculate the average. The data was converted back to polar form as seen in Table 1.

The graphs in the Figs. 7 and 8 show a comparison between the measured impedance values as a function of the theoretical values with the same data presented in Table 1. The graph of Fig. 7 presents the data in polar form while in Fig. 8 in rectangular form.

The Fig. 9 presents the 2100 values of all impedance measurements performed in the test without any form of data treatment where we can observe the clouds of impedance measurements compared with the theoretical values represented in polar form.



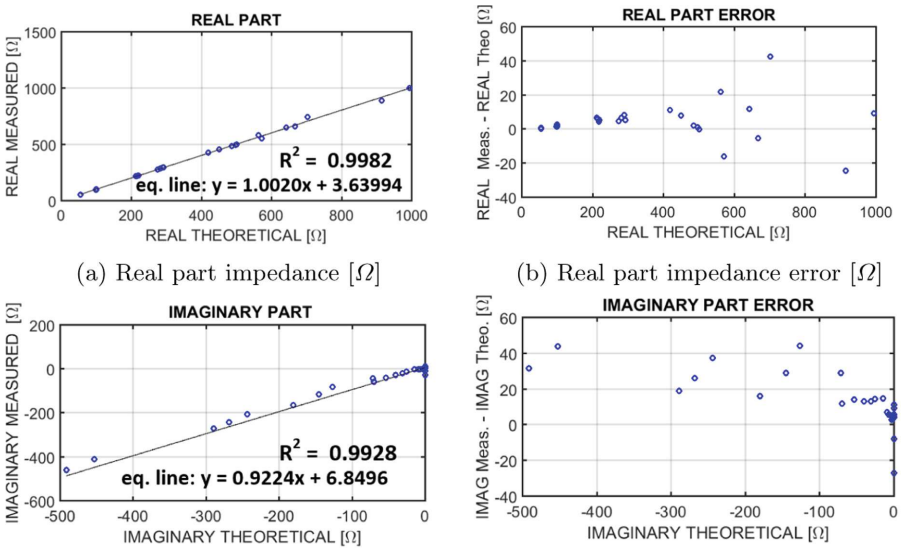
(a) Amplitude impedance [Ω]

(b) Amplitude error [Ω]

(c) Phase angle [°]

(d) Phase error angle [°]

Fig. 7. Impedance error in polar mode.



(a) Real part impedance [Ω]

(b) Real part impedance error [Ω]

(c) Imaginary part impedance [Ω]

(d) Imaginary part impedance error [Ω]

Fig. 8. Impedance error in rectangular mode.

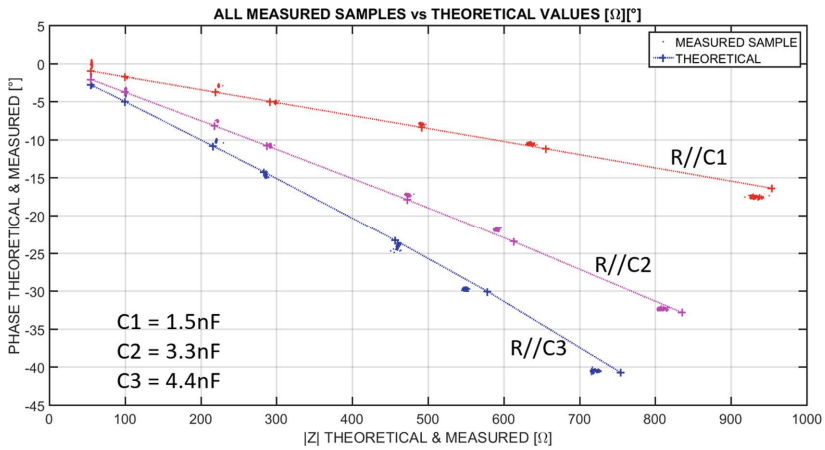


Fig. 9. All measured samples vs theoretical values

5 Discussion and Conclusion

Analyzing the measured and theoretical impedance results presented in the Figs. 7 and 8, we verified a correlation with Pearson's coefficient of $r = 0.9981 + 0.0134j$ indicating a positive linear relationship of the order of 99%. The amplitude of the standard deviation of the measured modules was $\pm 15.0175\,\Omega$, the average error was $\pm 1.8946\,\Omega$. Regarding the impedance phase of the measurement, the standard deviation found was $\pm 0.3458°$ and an average error of $\pm 0.2786°$.

The project is currently in its initial concept stage and only a prototype was built to analyze the effectiveness of the proposed architecture through experimental tests on phantoms. In the next stage, we will perform impedance measurements in biological tissues *ex-vivo* to compare these data with the impedance values known in the literature for the tissue under analysis. Then it is necessary to adapt the electrical insulation of the equipment to meet the safety requirements necessary for application in human beings.

Some changes must be made to achieve these goals. First, the volume of the impedance meter has to be reduced. A printed circuit board (PCB) is being developed to accommodate only the electronic components necessary for the project, avoiding the use of open source platforms and redundant electronic circuits. The concept of this modular architecture allows the implementation of multiple measurement channels. However, the maximum number of possible channels is 127, limited only by the Arduino's I^2C communication capability with the peripherals.

In summary, we believe that the impedance meter can be a viable alternative to contribute to studies related to electrical impedance myography.

A Quadrature Demodulation Algorithm

To determine the amplitude, phase and offset parameters, the algorithm developed by Zhang [19] known as quadrature demodulation is used. Quadrature demodulation is a high performance algorithm developed to determine the four parameters of a sine wave discretized in time in digitization processes. The algorithm is based on the iterative method of total least squares widely used in digital signal processing for linearization and function adjustment. However, quadrature demodulation is a non-iterative method that provides fast results with low computational load, reducing processing time. In this way, the amplitude, phase and offset of a sine wave can be easily determined by:

$$s = A\sin(\alpha+\beta) + c = A\sin(\alpha)\cos(\beta) + A\cos(\alpha)\sin(\beta) + c \tag{6}$$

By rearranging Eq. 6 in matrix form, we can previously calculate a E matrix with known points of unit amplitude sines and cosines, phase α and with the same sampling frequency (31.25 kHz) as the discretized signal, we have:

$$[s] = \begin{bmatrix}\sin(\alpha) & \cos(\alpha) & 1\end{bmatrix}_{N\times 3} \begin{bmatrix} A\cos(\beta) \\ A\sin(\beta) \\ c \end{bmatrix}_{3\times 1} = E \begin{bmatrix} a \\ b \\ c \end{bmatrix} = Ep \tag{7}$$

Then;

$$[s] = [E][p] \tag{8}$$

The pseudo-inverse of E can also be previously calculated because it is the matrix with the values of the known points of sines and cosines with unitary amplitude. Thus, we can find the coefficients of the matrix p by multiplying both sides of the Eq. 8 by the pseudo-inverse matrix $E^{\dagger}$, so we have:

$$[p] = [E^{\dagger}][s] \tag{9}$$

Then, the amplitude, phase and offset of the discretized sine wave are determined by the equations:

$$A = \sqrt{a^2 + b^2} \tag{10}$$

$$\theta = \arctan\frac{b}{a} \tag{11}$$

$$c = \mathit{offset} \tag{12}$$

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